

Options Lab — Chain Replay (design proposal)

Status: Design settled (Coach ticks §9, 2026-08-26). Not a Specification. Not BUILD AUTHORITY.

Date: 2026-08-26

Name: Chain Replay (Coach 2026-08-26)

Next: technical Spec, then a full-agent build plan — after this design is settled.

Type: Product / interaction design — instant replay of **OPF-held chain generations** already in the member's Heatmap **Cache** (TR14 stream book).

This document is the design we work out before a Spec. It does not replace Time Machine Spec v0.1.7, Heatmap Templates, OPF, or TR14.

0. Coach intent (do not drop)

Verbatim from this thread, in order:

1. The cache is a store of the **raw data coming from OPF as a chain** that is used to construct **all** the possible views created by Heatmap templates.
2. Theoretically, play back the cache and get an **instant replay of any template**.
3. If it is the raw data, then it should be **applicable to any template**.
4. It should also be able to **reconstruct a chart in the Analyzer** — e.g. construct a butterfly over the last so many minutes of cached raw data.
5. **Instant replay of heatmaps and strategy positions**. In fact **any strategy position within the timeframe of the cached data**.
6. In the Heatmap or the Analyzer, **substitute the live data with the cached data**, and create a **runner tool** that would allow the user to **scrub through the data**.
7. A data scrubber **similar to the Time Machine**, with a **range equal to the cached data**.
8. Even at **10 second intervals realtime playback is not terrible**.
9. Give the user the option of capturing **max detail for so many minutes** or **decimated detail for longer playback**. Put the cache in terms of **playback time instead of MB**, with a note that **granularity degrades as time to replay increases**. A slider that makes that relationship clear.
10. This would have to be a **setting once made, it would be going forward**. Any cached data after making the choice **might be destroyed or altered**.

11. Optimally they would **set the slider in the morning**, then they would have a **trailing replay starting at market open**. Or whenever they **reset the slider the trail would start from there**.
 12. If you are scrubbing, you are **not concerned about realtime accuracy**.
 13. **Start with the design**. Once details are worked out: technical spec, then build plan.
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1. What it is

Chain Replay is **OnDemand for the chain you already watched this tab** — not a downloaded day from gold disk.

The member is already subscribed to one OPF book (symbol + listed expiration + wing window). Live ticks paint Heatmap / Analyzer as today. Those ticks are **also recorded** into the Heatmap Cache (TR14 stream book): raw listed chain generations, this browser tab, RAM, member budget.

Replay **does not change templates or structures**. It **swaps the input**:

Mode	Input stream	What you see
Live	Current OPF generation (WebSocket)	Now
Replay	A selected slot in the tab cache	That as_of

Any Heatmap template (Advanced flies, Verticals, GEX, Width Fit, ...) and any **strategy position whose legs are listed on that cached chain** can be rebuilt at that tick. Same film, different projector.

It is **not** Time Machine. Time Machine remains: pick a **calendar day**, **download** a 390-minute underlier path, play **price and time** of that day (gold / server fetch). Chain Replay's film is **this tab's OPF chain ring**. Different SoR, similar **video grammar**.

2. Why Time Machine chrome, not Time Machine film

Members already know: play / pause / stop, a small window, a **draggable scrubber**, a clock, a range, a blue inner-edge when in replay. Reuse that **grammar** so Replay does not invent a second video language.

The as-built Time Machine stack is already split into **transport** (clock math + buttons + HUD) and **film** (downloaded 390-minute underlier path). Chain Replay keeps the transport. It swaps the film.

2.1 Four clocks (do not collide names)

Analyzer already has seats. This design adds one **without stealing Time Machine’s name**.

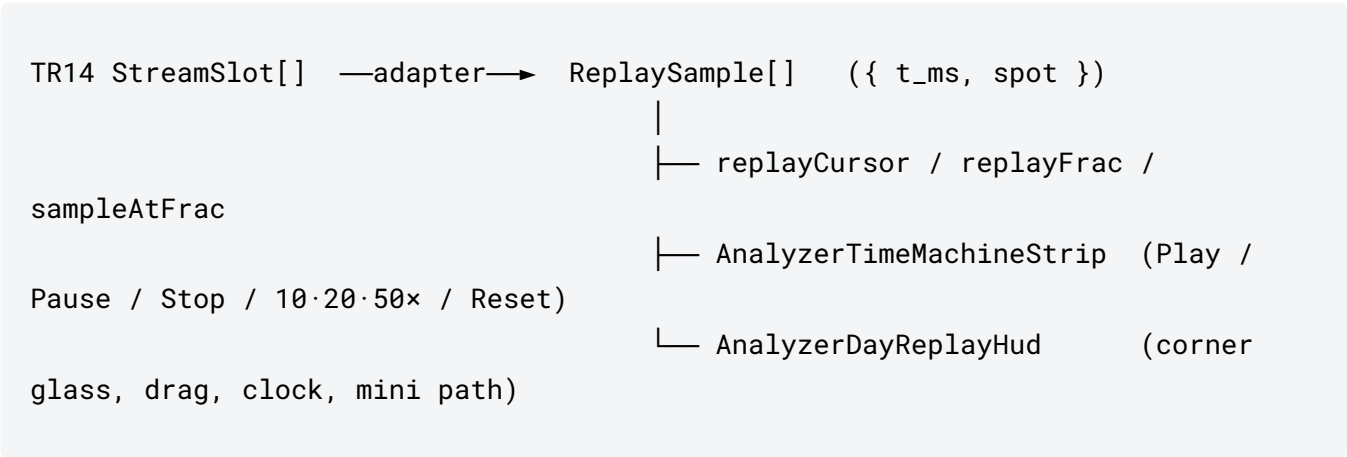
Seat	Job	Film / clock	Glow
What-if	Ad-hoc time · spot · vol knobs	Member sliders on the live/held sheet	Red inner
Time Machine	Pick a calendar day , download, play	Gold / server 1-minute underlier path (390 closes or TPO)	Blue inner
Chain Replay (working name)	Scrub this tab’s OPF chain ring	TR14 StreamSlot[] already in RAM	Same blue inner as TM (replay-on tell)
Surface Time machine	3D snap-rebind of listed-leg IV	Surface App Spec — not this	—

Time Machine Spec **ATM-B2** turns GEX / Probability **off** because that film is **underlier-only** — there is no chain at (t). Chain Replay **has** the listed chain at each tick, so Heatmap GEX (and any other template) is honest on the selected gen. That is why we do **not** inherit Basic TM’s GEX lock.

Mutual exclusion (design law): Time Machine and Chain Replay cannot both drive the Analyzer canvas. Entering one **Resets** the other. What-if may still overlay Chain Replay as ad-hoc σ/τ (same as TM **ATM-B4**); Spec can lock this. Surface TM stays a different product.

2.2 Reuse the transport as-built (no second video clock)

The Time Machine clock already walks an array of { t_ms, spot }. A cached generation already has asOf and spot. The adapter is one function; the rest of the stack is unchanged.



Adapter (new, small): slotsToReplaySamples(slots): ReplaySample[]

StreamSlot field	ReplaySample field	Rule
asOf (ISO) or receivedAt	t_ms	Prefer listed asOf ; fall back to receive time; drop slots with no usable time
spot	spot	Skip or keep last-named if null — Spec; design default: omit gens with no spot from the mini path, still selectable by hash if the host needs them
contentHash / index	(host, not HUD)	Host maps cursor.idx → slot → run()

StreamBook.atTime(key, tMs) **already exists** — nearest cached gen to a playhead. Chain Replay’s host should use it (or cursor.idx into window()) instead of a new lookup.

Reuse (as-built Time Machine tools)

Existing piece	Path	How Chain Replay uses it
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Play / Pause / Stop	<code>AnalyzerTimeMachineStrip.tsx</code> — <code>IconPlay</code> / <code>IconPause</code> / <code>IconStop</code>	Same hits, same labels. Hide the calendar <code><input type="date"></code> (optional prop). Range is the book, not a picked day.
Speeds 10× / 20× / 50×	<code>REPLAY_SPEEDS</code> in <code>algoDayReplay.ts</code>	Same detents. <code>replayCursor</code> already multiplies wall elapsed by speed and binary-searches <code>t_ms</code> . 10s film at 10× is still readable. Do not fork a second cursor.
Seek math	<code>replayCursor</code> / <code>replayFrac</code> / <code>sampleAtFrac</code>	Unchanged if the adapter emits <code>ReplaySample[]</code> . Clock stays America/New_York.
Mini HUD + scrubber	<code>AnalyzerDayReplayHud.tsx</code>	Same corner window (dark glass, drag, sky path, playhead line). Caption changes: day label → book span (10:32–10:51) or “Trail”; <code>N closes</code> → <code>N gens · 10s</code> . Extract shared caption props rather than copying the HUD.
Blue inner glow	Analyzer viewport when Time Machine is on	Same replay-on tell on Heatmap workspace and Analyzer canvas. What-if red glow unchanged. Both replay seats share blue so “I am not live” is one color.
Reset	Time Machine Reset (v0.1.7, ATM-S3)	Reset leaves Replay: paint returns to Live , HUD hides, blue glow off. Does not wipe the book (recording continues). Analogous to leaving the day, not deleting gold.

Stop	ATM-S2	Stop parks at the first stored gen (start of the trail), does not exit Replay. Same as TM Stop → first print.
HI DetentSlider	Cache slider today (MB stops)	Repurpose: playback-time stops, valuetext shows interval (~20 min · 2s).

Component surgery (prefer extract, not copy)

1. `ReplayTransportStrip` — Play / Pause / Stop / speeds / Reset. Time Machine wraps it and **adds** the date field on the left. Chain Replay wraps it and **does not**. Today's `AnalyzerTimeMachineStrip` is that wrap plus the date.
2. `ReplayHud` — today's `AnalyzerDayReplayHud` with `title` and `footer` props (`Trail / 47 gens · 18 min · 10s` instead of `Day / 390 closes`).
3. `algoDayReplay.ts` — no second clock module. Optional: `ReplaySample` may grow an optional `hash?: string` later; v1 host keeps `idx→slot` beside the samples.

Heatmap does not have this strip today. **Same components**, hosted on the Heatmap workspace (HUD over the grid; strip in the inspector Cache section or a thin bar above the grid). Do not invent Heatmap-only video buttons.

Do not reuse

Time Machine piece	Why not
Day calendar / download	No gold fetch. Range is the book.
390 candles / close-to-close vs TPO	Different film (underlier minutes vs chain hashes). Mini path is spot-from-gen , one vertex per slot, not OHLC candles.
Autofit X = session open (ATM-O1)	Replay X is the spot on the selected gen , or live Autofit when Live.

Basic TM: GEX and Probability off (ATM-B2)	Chain Replay may show GEX — the chain is in the slot. Analyzer GEX/Prob during Replay is a later Spec choice; default design: allow Heatmap GEX on scrubbed gens.
WAITING / NO PATH / NO MARKS (ATM-B5)	Named holes stay, different copy: NO FILM (empty book) · WAITING (first gen not yet) · NOT TRADED / CHECK LEGS on legs. Do not say NO PATH (that is a missing gold day).
“Add the fly after the day, then Algo” (ATM-A1)	Algo-on-gold-day stays Time Machine. Chain Replay may later drive a package mark path ; do not silently point Algo at RAM gens without a Spec line.
Server day fetch / OnDemand	Never. One market WebSocket already subscribed.

Time Machine’s **later vol plane** (FI-036 — full chain snaps from gold / StudioOne) is the **cousin** of this feature: same *kind* of film, different SoR (disk day vs tab RAM). This design does not implement that plane. A future Spec can feed gold chain snaps into the **same** adapter + HUD.

3. The film (cache)

SoR: TR14 stream book — `getStreamBook()` — this tab’s RAM.

Each slot: raw OPF generation already subscribed: listed dual-side contracts, spot, wings, strike step, `content_hash`, `as_of`, stale/quality. Optional Width Fit color memo is a **shortcut**, not a second store of truth. Templates stay **pure** (TR5): they never read the book; the **host** selects a slot and `run()` s.

Write (as-built + this design): Heatmap records a slot when a generation lands, **any** template (not Width Fit only). Analyzer in the **same tab** may read the same singleton after SPA navigation; a reload or new tab is empty.

Budget: internal ceiling remains 4 / 8 / 16 / 32 MiB. The member does **not** see megabytes as the primary control.

Live vs record: Live paint stays on the socket (2s class). The book **samples on write** at the interval implied by the playback slider. Scrubbing does not require 2s accuracy.

Gaps stay gaps. Same hash → one slot. Skipped seconds (decimation) are named on the playhead, not interpolated.

4. Playback-time slider (replaces “Cache MB” as the member story)

Job: set **how long I want to be able to scrub**, going **forward**. Granularity is the consequence.

Planning detents (busy ODTE, ~50–80 KB/gen, ~32 MB ceiling — tune in Spec):

Member-facing stop	Sample interval	Feel
~15–20 min	~2s	Max detail (what we collect)
~30–45 min	~5s	Coarser
~60 min	~10s	Longer film; 10s steps are acceptable playback

Copy on the control (Tango/Echo in Spec): **Longer playback uses fewer snapshots. Detail gets coarser.**

Valuetext example: ~20 min · 2s steps / ~60 min · 10s steps.

Once chosen, going forward only. Changing the slider **cannot** invent 2s ticks from a 10s film.

On change (design law):

1. Confirm (or a single destructive detent with a caption): replay from here uses this setting; **what's already cached will be cleared**.
2. Wipe the book (preferred v1 — clean trail).
3. Trail **starts now** (or at the next gen). Same as a new morning.

Morning ritual: set the slider before or at the open → first gen after open starts the film. First D minutes **fill**; after that it is a **trailing** window of length D . At 11:00 with a 20-minute max-detail setting the member has ~**10:40–11:00**, not 9:30–11:00. Resetting the slider starts a **new** trail from that moment.

8 MB today is ~3–5 min of max-detail ODTE — too short for “hour replay.” The slider **uses the 32 MB ceiling** (or the highest detent needed) as an internal cap; if a stop cannot fit, the stop is omitted or the interval coarsens further. Spec will freeze the stop list. Do not show a 6-hour stop that does not fit.

4a. Member walkthrough (the design in use)

Morning (or whenever)

- Heatmap inspector: set playback-time slider (~20 min · 2s or ~60 min · 10s)
- confirm wipe if a film already exists
- live OPF paints; host records sampled gens into TR14

Watch live

- any template (flies, Verticals, GEX, Width Fit, ...)
- Cache line: “47 gens · 18 min · 2s” (span grows until D , then trails)

Enter Replay

- Live | Replay (or Width Fit Live | Average | Replay)
- blue inner glow
- Time Machine-class strip + HUD appear
- playhead on last gen (or first – Spec); member scrubs or hits Play

Scrub / play

- tiles, GEX, Width Fit Live cell, Verticals – all re-run() on the selected slot

```
→ switch template without leaving Replay (same film)
→ SPA to Analyzer in this tab: same book, same playhead; tent / package
mark at that gen
→ missing listed leg → named state, never a fake debit
```

Reset

```
→ paint back to Live; HUD gone; blue off; book still recording
```

The member never picks a calendar day. They never wait for a download. If they never opened Heatmap this tab, Analyzer says **NO FILM** — not a gold fetch.

5. Runner / scrubber (the tool)

One runner tool, two hosts.

```
Live OPF → host records into TR14 book (sampled)
          |
          ├── Live mode: templates/sheet read current gen
          └── Replay mode: playhead selects a slot → same templates/
sheet run()
```

Range = first stored `as_of` → last stored `as_of` (and gen count). Never a clock longer than the book.

Controls (Time Machine grammar):

- Play / Pause / Stop
- Speed 10× / 20× / 50×
- Draggable scrubber on a mini path
- Clock: session time of the selected gen + step size (`10:42:00 · 10s`)
- Optional: `n gens · span` (e.g. `47 gens · 18 min`)

While Replay is on: Live socket **keeps recording** into the book (trail still grows; playhead does not have to stick to live). Pause is **paint-only** — it does not freeze the book. Stop parks at the first stored

gen. Reset returns **paint** to Live (book still recording).

Heatmap host: tiles, GEX profile, Width Fit. Average remains a **mean of a window**; Replay is a **single gen** under the playhead. Do not collapse Average and Replay. Coach: **Live | Average | Replay** on Width Fit; **Live | Replay** on other templates.

Analyzer host: substitute the chain used to **package-mark** representable legs with the selected gen. A butterfly whose three strikes are on that chain shows listed debit/credit (and listed greeks if present) **at that tick**. Missing leg → named state (NOT TRADED / CHECK LEGS), never a fake tent. The Risk graph vs **spot** is still a model sheet at that tick's spot; the **time path** of the package is the scrubber (mini HUD). This is **not** Time Machine's 390-minute underlier walk unless the film is gold.

6. Heatmap vs Analyzer (where chrome lives)

	Heatmap	Analyzer
Record	Yes — this is where the book is filled today	Same tab may read the book; recording stays Heatmap-owned in v1 unless Analyzer also subscribes the same book
Playback-time slider	Inspector Cache section, relabeled to time	Do not duplicate the slider; Analyzer Replay uses whatever Heatmap already recorded in this tab
Scrubber HUD	Shared <code>ReplayHud</code> (today's <code>AnalyzerDayReplayHud</code> + caption props) over the grid	Shared <code>ReplayHud</code> + <code>ReplayTransportStrip</code> ; hide the day calendar ; TM date field stays on Time Machine only
Enter Replay	Host control (e.g. next to Cache, or Live/Replay)	Viewport strip: Replay on if the book has gens; else NO FILM . Entering Replay Resets Time Machine if it was on

Cross-route	SPA same tab: singleton survives Heatmap → Analyzer. Full reload: empty	If the member never opened Heatmap this tab, Analyzer Replay is NO FILM — tell them, don't fetch gold
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Heatmap is the recorder. Analyzer is a **second projector** on the same tab's film. Opening Analyzer first does not create a chain ring (v1). Analyzer Replay with an empty book is **NO FILM**.

7. Honesty (non-negotiable)

- OPF-held listed chain only. No invented strikes, mids, or package prices (DL-309 / OT-EF).
- Representable or **named** failure at that tick.
- Decimation and slider-clear are **named** (interval on the clock; confirm on slider change).
- Process / inspection only. No profit claims on replay.
- One market WebSocket. Replay does not add a socket or a Massive path.
- Not gold disk. Not a full session day at 2s. Not MiniTwo server cache.

8. What already exists vs this design

Exists	This design
TR14 book, 4–32 MB detent, drop-oldest	Member story = playback time ; interval derived
Write on Width Fit only (until DL-593)	Write any Heatmap template (as-built now)
Width Fit Average	Unchanged; Replay is a different host view
Time Machine strip + HUD + replayCursor	Same transport ; adapter StreamSlot[] → ReplaySample[]; hide date
<code>StreamBook.atTime</code> / <code>window</code> / <code>clear</code>	Playhead lookup, trail window, slider-wipe

Analyzer package quote on live/held chain	Package quote on selected cached gen when Replay
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9. Coach ticks (2026-08-26)

#	Question	Coach
1	Product name	Chain Replay. Time Machine stays the calendar-day download.
2	Width Fit modes	Live Average Replay. Average = mean of a window. Replay = single gen under the playhead. Other templates: Live Replay.
3	Record while scrubbing	Yes. Socket still fills the book. Pause is paint-only.
4	Slider change	Wipe the book + confirm. Going forward only. Trail starts now.
5	Who records (v1)	Heatmap records; Analyzer projects. Analyzer with no Heatmap film this tab → NO FILM.
6	Algo Alert in v1	Out. Time Machine keeps Algo-on-day. Chain Replay is inspection, not a trail bot.
7	Speeds	Keep TM 10× / 20× / 50×. Clock shows step size. No extra 1× in v1.
8	What-if overlay	Allow. Replay owns the gen's spot and session time; What-if may still apply ad-hoc σ/τ (same as TM ATM-B4).

§9 is **settled**. Next document is the technical Spec when Coach opens it.

10. Out of this design

- Gold-disk / StudioOne full-day chain replay (that can **feed** Time Machine later; it is not this RAM ring)
 - Server-side member cache
 - Multi-tab or multi-device film
 - Upsampling 10s slots to 2s
 - Second Massive pipe
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11. Suggested path after this document

1. ~~Coach names the feature and ticks §9.~~ **Done 2026-08-26.**
2. **Specification** (Heatmap + Analyzer host, TR14 amendment, TM chrome mapping, ATs).
3. **Full-agent bench plan** (Echo labels → Charlie/Alpha → Kilo → Delta).

No implementation until that Spec has BUILD AUTHORITY / GO.

Key decisions (design, pending Coach)

Decision	Rationale
Film = TR14 tab RAM, not gold	Instant; already subscribed; minutes-to-an-hour, not 390
Swap input stream , keep templates	Pure templates; any view; any representable position
Time Machine transport , not TM download	Adapter into existing <code>ReplaySample</code> / cursor / HUD / strip; names stay distinct
TM and Chain Replay exclusive	One playhead on the Analyzer canvas

Heatmap GEX on during Replay	Film includes the chain; Basic TM GEX-off does not apply
Slider = playback time ; MB internal	Time is the job; granularity is the cost
Going forward + wipe on change	Cannot invent finer ticks; trail starts at set/reset
Heatmap records; Analyzer may project	One book; Analyzer empty until Heatmap has filled this tab
Live keeps painting 2s; book samples	Scrub does not need realtime accuracy
Range = cached gens only	Never a longer clock than the book

One-line: Live OPF paints now; this tab records a trailing chain film whose length you set in minutes; Replay uses Time Machine's play/scrub chrome to walk that film on any Heatmap template or any listed-leg position — never a second market, never a fake print.